Deep Learning-based Multi-modal MRI Automatic Segmentation of Gliomas

I. Background

Gliomas are the most common primary intracranial tumors found in clinical practice. Doctors typically utilize various MRI modalities, specifically T1, T1ce, T2, and FLAIR, to identify distinct tumor regions, such as the enhancing tumor zone, the tumor core, and the whole tumor area. Manual segmentation is not only time-consuming but also prone to subjective differences between practitioners. This project aims to use the BraTS (Brain Tumor Segmentation) dataset to develop an automated deep learning-based segmentation algorithm.

II. Dataset

Public BraTS dataset in NIfTI format (.nii.gz) can be accessed on the 云桌面 (Home/Downloads/dataset) along with demo codes (Home/Desktop) for reading the data.

Modalities: Each patient record includes four co-registered 3D modalities plus one Ground Truth label.

Label Definitions:

Label 1: Necrotic/Non-enhancing Tumor.

Label 2: Peritumoral Edema.

Label 4: GD-enhancing Tumor.

III. Project Assignments

Task 1: Data Preprocessing and Multi-dimensional Visualization (Level: Easy)

Objective: Familiarize yourself with medical imaging formats and master the statistical characteristics and preprocessing workflows of multi-modal data.

Image Reading: read .nii.gz files and convert them into Numpy arrays.

Visualization: Use Matplotlib to display the four modalities of the same patient at identical slice positions and overlay the Ground Truth masks.

Preprocessing Pipeline:

Implement Z-score normalization (subtract the mean and divide by the standard deviation).

Implement Data Cropping to remove redundant black background areas and reduce computational load.

Label Transformation: Convert the original labels (1, 2, 4) into the three nested regions of clinical interest: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

Contrast Analysis: Calculate and compare the mean pixel intensity of the tumor region versus the surrounding healthy brain tissue across all four modalities. Which modality provides the best contrast for the "Enhancing Tumor" region?

Deliverables: Example visualization results and the processed 2D slice dataset.

Task 2: Baseline Model Construction and 2D Segmentation Training (Level: Moderate)

Objective: Independently implement the basic network architecture and master loss functions and training strategies for segmentation tasks.

Architecture Implementation: Use PyTorch or TensorFlow to build a U-Net (or a similar Encoder-Decoder architecture).

Loss Function Design: Address the medical image class imbalance (where the tumor area is much smaller than the background) by implementing Dice Loss or a Dice + Cross Entropy hybrid loss function.

Data Augmentation: Apply random rotation, flipping, scaling, and Gaussian noise to improve model generalization.

Model Training: Slice the 3D data into 2D images for training. Set a reasonable Batch Size and learning rate decay strategy (e.g., ReduceLROnPlateau).

Preliminary Evaluation: Calculate the Dice coefficient for each region on the validation set.

Deliverables: Model definition code, training logs (Loss curves), and prediction maps of preliminary segmentation results.

Task 3: Performance Optimization, Advanced Models, and Clinical Indicator Analysis (Level: Challenging)

Objective: Introduce advanced architectures to solve spatial continuity issues and conduct a comprehensive evaluation using clinical standards.

Architecture Upgrade (potential directions to be explored are listed below):

- Upgrade the network architecture to utilize inter-slice spatial information.
- Introduce Attention U-Net (attention mechanisms) to improve the detection of small lesions.

Advanced Quantitative Metrics: In addition to the Dice coefficient, calculate the Hausdorff Distance (HD95) to evaluate boundary segmentation accuracy.

Error Analysis: Identify the 5 worst-performing cases, analyze whether the failure was due to low contrast, small lesion size, or artifacts, and propose improvement ideas.

Deliverables: A complete project code repository and a final experimental report.

IV. Submission Guidelines

Deadline (week 15): Presentation, June 8th; Report, June 14th

The code, report (in PDF format) and presentation slides need to be uploaded to the E-learning system before the deadlines. Failure to submit your code will result in a 40% reduction of the total assignment mark.

Only **one** group member is required to submit the required materials for the entire team. **Remember to list the Chinese names of all group members in the slides and report.**

V. Grading Policy

Evaluation Item	Weight	Evaluation Criteria
Data Processing (Task 1)	15%	Clear code logic, correct preprocessing methods, and high-quality visualizations
Model Implementation (Task 2)	30%	Correct network structure, appropriate loss function selection, and successful model convergence
Performance Breakthrough (Task 3)	20%	Improved Dice metrics with clear optimization attempts
Report	15%	Complete charts, rigorous analysis of metrics, and insightful discussion.
Presentation	20%	Clear introduction of the group division of labor, methods, innovations, and results discussion and **use of AI in this project.**